

Limb-sparing surgery

Limb-sparing (or limb-salvage) surgery is the most common type of surgery for a primary bone cancer or a soft tissue sarcoma in an arm or leg.

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About limb-sparing surgery

This is the most common type of surgery for a [primary bone cancer](#) or a [soft tissue sarcoma](#) in an arm or leg. It is sometimes called limb-salvage surgery.

For a soft tissue sarcoma, the surgeon removes the tumour and an area of healthy tissue around it. You might need some [reconstruction surgery](#) to repair the area.

For a bone cancer the surgeon removes the affected part of the bone and may replace it with either:

- a metal implant (prosthesis)
- bone from another part of the body, or from someone else (bone graft).

If the cancer affects a bone in or near to a joint, the surgeon can often replace the joint with an artificial one.

Preparing for limb-sparing surgery

Before the operation, your surgeon will explain:

- what is involved in the surgery
- how the surgery might affect you
- the possible risks and complications of the operation.

You can ask your surgeon how the surgery might affect you.

If you have a bone cancer and are having limb-sparing surgery with a metal implant, you might like to see the implant before it is fitted. Your [surgeon, specialist nurse or keyworker](#) can usually arrange this for you. There is a risk the metal implant could become infected. Your cancer doctor will explain more about this and what they will do to reduce the risk of infection.

You will meet some of the rehabilitation team who will help you after your operation. They will help you keep as much movement as possible in the affected limb. This is so you can be as independent as possible after the surgery. They can also give you information about your recovery.

The team will include specialist doctors and nurses, as well as:

- a physiotherapist
- an occupational therapist (OT)
- a social worker.

A physiotherapist may talk to you before your operation. They may give you some exercises to do before surgery, to help strengthen your muscles.

You may find it helpful to talk to someone who has had the same operation. Your healthcare team or keyworker may be able to arrange this for you. Some hospitals have a counsellor you can talk to about your [feelings](#) before the operation. You can also talk to them about any worries about a [change in your appearance](#).

You may also find it helpful to contact [Sarcoma UK](#) or the [Bone Cancer Research Trust](#).

After your limb-salvage operation

When you wake up from the operation, you may have a drip (intravenous infusion) going into your arm to replace your body's fluids. This can be removed as soon as you are drinking enough.

Your wound

You may have some tubes (drains) to drain fluid from the wound. These are removed when fluid has stopped draining. This should take a few days.

Sometimes, fluid can build up around the wound. The swelling should reduce over a few weeks. Sometimes, a lot of fluid builds up around the wound. This is known as a seroma. It may need to be drained by a doctor or nurse.

Your surgeon may use stitches or clips to close the wound. Sometimes they use dissolvable stitches. If you have stitches or clips that need to be removed, you will be told when and where this will happen.

The nurses will monitor the wound for any signs of infection. Let them know straight away if:

- you feel unwell and have a high temperature
- there is any warmth, swelling and redness around the wound
-

there is discharge from the wound.

Pain

Pain can usually be controlled effectively with painkillers. They may be given:

- into a vein (intravenously)
- into the space around your spinal cord (epidural)
- into a muscle (intramuscularly)
- as tablets.

To start with, you will probably need a strong painkiller, such as morphine.

You may be given intravenous pain relief through a syringe connected to an electronic pump. The pump can be set to give you a continuous dose of painkiller.

You may also have a handset with a button you can press if you feel sore. This is called patient-controlled analgesia (PCA). It is designed so that you cannot have too much painkiller (an overdose). So it is okay to press it whenever you are uncomfortable.

If you have had surgery to your leg, you may have pain relief using an epidural. This is a fine tube inserted through your back into the area around the spinal cord (the epidural space). A local anaesthetic can be given continuously into this space to numb the nerves that run to the legs.

Let your nurses and doctors know as soon as possible if you are in [pain](#). This will help them give you the combination and dose of painkillers that is right for you.

Exercises

Your nurses or the physiotherapist will teach you breathing and leg exercises. You can help yourself get better by doing the exercises as often as you need to. They will also encourage you to move around as soon as possible.

Breathing exercises will help prevent you getting a chest infection. Leg exercises will help stop blood clots forming in your legs. Chest infections or blood clots can develop if you are not moving around as much as normal.

To help prevent a blood clot in your leg (a deep vein thrombosis or DVT), you may be given elastic stockings (TED stockings) to wear. You may also be given injections of anti-clotting medicines to help thin your blood.

Rehabilitation

Limb-salvage surgery is a major operation and rehabilitation can sometimes take a long time. It can often be hard work and may be frustrating at times. This will depend on which limb is affected and how big the operation was.

You will see a physiotherapist soon after your operation. They will encourage and help you to move around as soon as possible.

The physiotherapist will show you exercises to help you keep the most function in the limb. It is important to follow their advice and regularly do the exercises. This will help you recover and reduce the risk of any problems after surgery.

The rehabilitation team will help you get as much independence as possible after the surgery. You may need to make changes to your normal activities. This will depend on the extent of the surgery and how it is affecting you. The occupational therapist will help you with any changes.

Your rehabilitation may take longer if you are also having chemotherapy or radiotherapy.

After surgery for soft tissue sarcoma

Sometimes, you will need a brace or support for your limb. If the operation was on your leg, you may be given crutches or another walking aid to use while you are learning to walk again.

After surgery for bone cancer

At first, your limb will be firmly bandaged, or splinted, to keep it still. This will give the bone graft or implant time to start joining firmly on to the rest of the bone in the limb.

If you have surgery to your hip or knee, you may have to wear a splint or brace for a few months. After surgery to your arm or shoulder, you may need to wear a sling for a few months. Your cancer doctor, specialist nurse or physiotherapist can tell you more about this.

If you have had an operation on your leg or pelvis, you will need help moving around at first. You will use walking aids to help you, such as a frame or elbow crutches. A physiotherapist will teach you how to get in and out of bed. They will also teach you how to safely use any walking aids.

Going home

How long you need to stay in hospital after limb-sparing surgery will depend on the operation you had. Your cancer doctor or specialist nurse can give you more information. You may need to stay in hospital longer to rehabilitate. This will help you cope better when you go home.

If you think you might have problems when you go home, tell your nurse or doctor as soon as possible. They can arrange help for you at home before you leave hospital. You may be referred to a community occupational therapist who can do a home assessment. This can help them see if you need any equipment to help you stay independent at home.

When you are at home, you will keep seeing a physiotherapist as an outpatient. This can often happen locally to where you live.

Body image

Limb-sparing surgery may cause a change in how your limb looks or works. This may affect how you think and feel about your body ([body image](#)). This usually improves as you adapt to any change. But if you still have concerns, talk to your nurse or doctor.

Lymphoedema

After surgery, some people may develop swelling called [lymphoedema](#). It happens because the lymph nodes and vessels can be damaged by surgery or radiotherapy.

Let your doctor know about any swelling you have after surgery. They can explain how it can be managed.

Replacement joints and bone grafts

After limb-sparing surgery, you can usually do most of the things you could before treatment. This includes exercise and sports. But there may be some things you cannot do, or you find hard. Your surgeon or physiotherapist can give you more advice about this.

Knee joint



When the knee joint is replaced, the new joint normally works very well. You will be able to do low-impact exercise such as swimming or riding a bike. But your cancer doctor might tell you not to do high-impact sports like hockey, football, trampolining or rugby. This is because of the risk of damaging or loosening the joint.

Sometimes, years after surgery, a prosthetic knee joint may loosen or cause pain. If this happens, you will need more surgery.

Hip joint



Replacing a hip joint is usually very successful. It can take time to recover from surgery and get your strength back. Your surgeon or specialist nurse can tell you how long this might take. At first, you will need to use crutches to walk. You may need to use a walking stick for a few weeks or months after surgery. If you do not have strong joints and muscles, you may need to keep using a walking stick for longer.

Hip replacements may loosen over time. Some people need more surgery a few years after having their hip replaced.

Shoulder joint



If you have a shoulder joint replacement, the aim of the surgery is to be able to move your arm normally below shoulder height. But you will probably not be able to raise your arm above shoulder height.

It is uncommon for a shoulder joint prosthesis to loosen. They should last for many years and do not cause many problems.

Bone grafts



When limb-sparing surgery is done on the straight part of a bone, it may be replaced with a piece of bone taken from another area of the body. This is called a bone graft.

Occasionally, an infection develops in the replaced bone. This can happen days or weeks after the operation. Contact your hospital doctor or nurse straight away if you notice any redness or swelling in the area or develop a temperature. The earlier an infection is found, the easier it is to treat.

Driving after surgery for bone cancer

After bone surgery, you will not be able to drive until:

- your bones have fully healed
- you have enough muscle control to move your limb.

Your healthcare team can tell you more about this.

Blue Badge scheme

This scheme allows people with mobility problems to park closer to where they want to go. [Gov.uk](https://www.gov.uk) has more information about the Blue Badge scheme and how to apply. Or you can contact your local council for more information. A healthcare professional, welfare rights adviser or social worker can help with the application.

Date reviewed

Reviewed: 01 August 2022 | Next review: 01 August 2025

⚠ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help